Nanotechnology in Medicine

Speech for the presentation, slide by slide

Slide 1 - Nanotechnology in Medicine

Hello everyone, and welcome. Today I will show you how nanotechnology is changing medicine. A nanometer is one billionth of a meter, and at this tiny scale we can build tools that diagnose, treat and even repair the human body one molecule at a time. Over the next twelve slides we will look at why the nanoscale matters, the main tools, applications in treatment, diagnosis and regeneration, and finally the benefits and the risks.

Slide 2 - Why the nanoscale matters

Why is the nanoscale so special for medicine? Because biology itself works there. A DNA helix is about two nanometers wide, proteins measure between two and ten nanometers, and viruses between twenty and two hundred. Nanoparticles are the same size as these natural machines, so they can interact with them directly, and they can cross biological barriers that larger objects cannot. They also have a very high surface-to-volume ratio, which means more room to carry drug molecules and more sites to bind to a target.

Slide 3 - The nanocarrier toolbox

Let us look at the toolbox. Liposomes are tiny bubbles of fat that can hold drugs in their water core or in their membrane. Lipid nanoparticles protect fragile molecules such as RNA. Polymeric nanoparticles are biodegradable and release their cargo slowly. Gold and iron-oxide particles are used for imaging and for heating tissue. Finally, dendrimers are branched carriers, and quantum dots are fluorescent crystals used as bright labels for cells.

Slide 4 - Targeted drug delivery

The first big application is targeted drug delivery. The goal is simple: put the drug exactly where it is needed and nowhere else. With passive targeting, nanoparticles slip through the leaky blood vessels of a tumor and accumulate there, a phenomenon called the EPR effect. With active targeting, antibodies on the particle surface recognize receptors on diseased cells. Some carriers even release their drug only when triggered by acidity, heat or light. A real example is Doxil, a liposomal form of doxorubicin, which reduces the heart damage caused by the free drug.

Slide 5 - mRNA vaccines

Many people met nanomedicine through the COVID-19 vaccines. Messenger RNA is destroyed within minutes in the body, so it needs protection. Lipid nanoparticles, about eighty to one hundred nanometers wide, wrap the mRNA and deliver it into our cells. The cells read the instructions and produce a harmless viral protein, and the immune system learns to recognize it. The same platform is now being studied for influenza, for cancer vaccines and for gene editing.

Slide 6 - Fighting cancer

Nanotechnology also offers new ways to fight cancer. In photothermal therapy, gold nanoshells accumulate in a tumor and heat up when exposed to near-infrared light, destroying the cancer cells. In magnetic hyperthermia, iron-oxide particles are heated by an alternating magnetic field. Albumin-bound paclitaxel, sold as Abraxane, is approved for breast and pancreatic cancer. Hafnium oxide nanoparticles can also boost the effect of radiotherapy. In general, healthy tissue is spared more than with classic chemotherapy.

Slide 7 - Nano-diagnostics and imaging

Nanotechnology is equally powerful for diagnosis. Iron-oxide nanoparticles are used as MRI contrast agents to reveal lesions in the liver or lymph nodes. Quantum dots give bright and stable fluorescent tags for cells. Gold nanoparticles produce the colored line in lateral-flow tests, such as pregnancy tests. Nanoscale biosensors can detect a few molecules of a cancer marker in a drop of blood. Detecting disease earlier means better chances of survival.

Slide 8 - Regenerative medicine

Another field is regenerative medicine. Nano-scaffolds mimic the natural extracellular matrix and guide cells as they grow. Nano-hydroxyapatite is used in bone grafts and dental implants because it resembles real bone mineral. Aligned nanofibers can guide the regrowth of damaged nerves. Silver nanoparticle dressings protect wounds against infection, and nanoparticles can release growth factors slowly to support healing.

Slide 9 - Fighting infection and the road ahead

Looking at the frontier, silver and zinc-oxide nanoparticles can damage bacteria that resist antibiotics. Theranostic particles combine diagnosis and treatment in a single object. Nano-coatings on implants reduce rejection. Researchers build DNA origami robots that open only when they touch a target cell. And teams are designing particles that cross the blood-brain barrier, which could open the door to treating brain diseases.

Slide 10 - Key milestones

Here are some milestones. In 1959 the physicist Richard Feynman gave his famous talk, There is plenty of room at the bottom. In 1995 Doxil became the first approved nanomedicine. Abraxane followed in 2005. In 2018 Onpattro became the first approved drug using lipid nanoparticles to deliver siRNA. And in 2020 mRNA COVID-19 vaccines showed the world what nanomedicine can do at global scale.

Slide 11 - Advantages and disadvantages

Let us weigh the advantages and the disadvantages. The advantages are precise delivery with fewer side effects, lower drug doses, earlier and more sensitive diagnosis, protection of fragile drugs such as RNA, the possibility to combine imaging and therapy, and new ways to repair tissue. The disadvantages are that long-term toxicity is still unclear, particles may accumulate in the liver or spleen, they are hard to manufacture consistently, research and treatment costs are high, regulation and approval are complex, and there are ethical and privacy concerns. This is why careful safety studies are essential.

Slide 12 - Thank you

To conclude, nanomedicine gives us smaller tools and bigger hope for patients. It already protects us with vaccines, treats cancers and improves diagnosis, and the best is probably still to come, as long as we develop it safely and responsibly. Thank you very much for your attention. I am happy to answer your questions.